In **T 133/92** the question to be answered in examining novelty was whether the selection of the alkyl group as defined in claim 1 of the disputed patent had been made available to the public with regard to the disclosure of a prior document. By citing **T 666/89** (OJ 1993, 495), the patent proprietors contended that the legally correct approach for deciding selection novelty was identical or very similar to that employed in determining obviousness. In particular, they argued that in cases of overlapping ranges of compounds, a claim to a narrower range as compared with a broader prior art range was always selectively novel if it could be demonstrated that the narrow range was inventive over the broader range. However, the board observed that in the case cited the board had repeatedly emphasised that selection novelty was not different from any other type of novelty under Art. 52 and 54 EPC 1973, so that the proper approach was to consider availability in the light of a particular document. Thus the board found that a claimed group of compounds, essentially resulting from omitting those parts of a larger group of compounds which a skilled person would have immediately considered as being less interesting than the rest, could not be selectively novel. In addition, in the board's opinion, a skilled person would, having regard to these considerations, have seriously contemplated applying the technical teaching of this prior art document in the range of overlap.

6.2.3 Novelty of enantiomers

According to decision **T 296/87** (OJ 1990, 195), the description of racemates did not anticipate the novelty of the spatial configurations contained in them; racemates were described in the state of the art by means of expert interpretation of the structural formulae and scientific terms; as a result of the asymmetric carbon atom contained in the formula the substances concerned might occur in a plurality of conceivable spatial configurations (D and L enantiomers), but the latter were not by themselves revealed thereby in an individualised form. That methods exist to separate the racemate into enantiomers was something that should only be considered with respect to inventive step.

In **T 1048/92** the board observed that the fact that the disclosure of the prior document did not embrace more than two possible steric configurations did not take away the novelty of the specific one which was claimed in the application, because there was no unambiguous technical teaching directed to that configuration. The novelty of such an individual chemical configuration could only be denied if there was an unambiguous disclosure of this very configuration in the form of a technical teaching. It was thus not sufficient that the configuration in question belonged conceptually to a disclosed class of possible configurations without any pointer to the individual member.

In **T 1046/97** it was decided that the term "optically-active forms" could not be equated to an individualised disclosure of a specific enantiomer (see also **T 833/11**).

6.2.4 Achieving a higher degree of purity

In **T 990/96** (OJ 1998, 489) it had to be examined whether the feature in dispute, which in fact represented a **specific degree of chemical purity** (in particular diastereomeric purity), constituted a "new element" imparting novelty to the claimed subject-matter. The board stated that it was common general knowledge that any chemical compound obtained by a chemical reaction would normally contain impurities for various reasons and that it was not possible for thermodynamical reasons to obtain a compound which was - in the strict sense - completely pure, i.e. totally free of any impurity. It was, therefore, common practice for a person skilled in the art of preparative organic chemistry to (further) purify a compound obtained in a particular chemical manufacturing process according to the prevailing needs and requirements. Conventional methods for the purification of low molecular organic reaction products, which could normally be successfully applied in purification steps, were within common general knowledge. It followed that, in general, a document disclosing a low molecular chemical compound and its manufacture made this compound available to the public within the meaning of Art. 54 EPC 1973 in all grades of purity as desired by a person skilled in the art (**T 392/06**).

In **T 728/98** (OJ 2001, 319) the applicant (appellant) argued that the situation was such an exceptional one as mentioned in **T 990/96**. The claimed pharmaceutical composition differed from the state of the art because the particularly high purity level of the compound it contained could not be achieved by conventional methods. The board found, however, that the applicant, who bore the burden of proving this allegation, had not provided the necessary evidence. Where the claimed purity level of a low molecular chemical compound (here a terfenadine derivative) turns out to be successfully achieved by applying a conventional purification method on a reaction mixture disclosed in the prior art, an exceptional situation such as addressed in decision **T 990/96** does not exist. This would have required evidence that conventional methods could not achieve that purity level. Therefore the general rule applies that the level of purity of that low molecular compound cannot entail novelty. That general rule is valid also in the case of a product-by-process claim where that purity level is the inevitable result of the preparation process indicated in the claim.

In **T 112/00** the board considered a composition including a solvent having a purity greater than 99% to be new over a prior art composition containing such a solvent whose purity was not specified. The board stated that the claimed composition could be considered as a final product and the solvent as the starting material. As in **T 786/00**, novelty would be established by the defined purity of the starting material.

The issue in **T 803/01** was the novelty of a pharmaceutical composition which differed from prior-art compositions only in the degree of purity of one of its components. In the board's view each and every purification method, provided it was "conventional" but regardless of the extent of purification sought, was presumed to be automatically available to the public, and that in a fully enabling way, so as to amount to an effective novelty-destroying disclosure. As stated in **T 100/00** in this respect, the term "conventional" could only mean "conventional in view of the concrete technical context concerned". Therefore the question of whether the degree of purity for the polylactide required in claim 1 provided a new element over the prior art had to be assessed in the concrete technical context concerned.

In **T 142/06** the board noted that it followed from the considerations made in decision **T 990/96**, according to which a document disclosing a low molecular compound and its manufacture normally makes this compound allowable in all desired grades of purity, that the purity level of an organic compound is as such not an essential feature for the definition of this organic compound. However, in the case at issue it was evident that the content of chlorine ion of the claimed latex was an essential feature of the claimed latex, since, according to the patent in suit, only the latexes having this low level of chlorine ions enabled the production of films having the desired properties in terms of oxygen barrier properties and boil blushing properties. This implied that the claimed degree purity in terms of chlorine ion content could not be considered as an arbitrary degree of purity but that it amounted to a purposive selection. Thus, for this reason the considerations made in decision **T 990/96** and, by way of implication, in decision **T 803/01**, did not apply to the case before the board.

In **T 1085/13**, the board decided that a claim defining a compound as having a certain purity lacks novelty over a prior-art disclosure describing the same compound only if the prior art discloses the claimed purity at least implicitly, for example by way of a method for preparing said compound, the method inevitably resulting in the purity as claimed. Such a claim, however, does not lack novelty if the disclosure of the prior art needs to be supplemented, for example by suitable (further) purification methods allowing the skilled person to arrive at the claimed purity (see also **T 1914/15**). The question of whether such (further) purification methods for the prior-art compound are within the common general knowledge of those skilled in the art and, if applied, would result in the claimed purity, is not relevant to novelty, but is rather a matter to be considered in the assessment of inventive step. Further, the board was convinced that the rationale of **T 990/96** and **T 728/98** was not in line with **G 2/88** and **G 2/10**.

### 6.3. Selection of parameter ranges

#### 6.3.1 Selection from a broad range

The principles applied by the boards of appeal on novelty of selection inventions were initially developed in particular in **T 198/84** (OJ 1985, 209). They are summarised briefly in **T 279/89**, according to which a selection of a sub-range of numerical values from a broader range is new when each of the following criteria is satisfied:

(a) the selected sub-range should be narrow;

(b) the selected sub-range should be sufficiently far removed from the known range illustrated by means of examples;

(c) the selected area should not provide an arbitrary specimen from the prior art, i.e. not a mere embodiment of the prior description, but another invention (purposive selection).

Since 2019, the Guidelines state that only the first two criteria need to be fulfilled (G-VI, 8 (ii) – November 2019 version; see e.g. **T 261/15**).

The three postulates for the novelty of a selected sub-range are based on the premise that novelty is an absolute concept. It is therefore not sufficient merely for the wording of the definition of an invention to be different. What has to be established in the examination as to novelty is whether the state of the art is such as to make the **subject-matter** of the invention available to the skilled person in a technical teaching (**T 198/84**, OJ 1985, 209; see also **T 12/81**, OJ 1982, 296; **T 181/82**, OJ 1984, 401; **T 17/85**, OJ 1986, 406).

With reference to the third criterion, the board in **T 198/84** was of the opinion that this view of novelty really entailed more than just a formal delimitation vis-à-vis the state of the art. There would be delimitation only in respect of the wording of the definition of the invention, but not in respect of its content, if the selection were arbitrary, i.e. if the selected range only had the same properties and capabilities as the whole range, so that what had been selected was only an arbitrary specimen from the prior art. This was not the case if the effect of the selection, e.g. a substantial improvement in yield, occurred in all probability only within the selected range, but not over the whole known range (purposive selection). To prevent misunderstanding, the board emphasised, following **T 12/81** (OJ 1982, 296), that a sub-range singled out of a larger range was new not by virtue of a newly discovered effect occurring within it, but had to be new per se. An effect of this kind was not therefore a prerequisite for novelty; in view of the technical disparity, however, it permitted the inference that what was involved was not an arbitrarily chosen specimen from the prior art, i.e. not a mere embodiment of the prior description, but another invention (purposive selection).

With reference to the third criterion (criterion "(c)"), several decisions consider that purposive selection is relevant for assessing inventive step but not novelty (see **T 1233/05**, **T 1131/06**, **T 230/07**, **T 913/07**, **T 1130/09**, **T 2041/09**, **T 492/10**, **T 1948/10**, **T 423/12**, **T 378/12**, **T 1404/14**, **T 261/15**, some of which are reported below). However, see for example decision **T 66/12**, in which the board left it open as to whether the first two criteria were satisfied in light of its conclusion that the third criterion was not met (conclusion – lack of novelty). See also **T 673/12** and **T 2438/13**, which recalled the three criteria for the examination of novelty.

In **T 17/85** (OJ 1986, 406) the novelty of the claimed range was denied because the preferred numerical range in a citation in part anticipated the range claimed in the application. A claimed range could not be regarded as novel, at least in cases where the values in the examples given in the citation lay just outside the claimed range and taught the skilled person that it was possible to use the whole of this range (see also **T 847/07** on prior art pursuant to Art. 54(3) EPC).

In deciding the question of the novelty of an invention, the board in **T 247/91** emphasised that consideration had to be given not only to the examples but also to whether the disclosure of a prior art document as a whole was such as to make available to the skilled person as a technical teaching the subject-matter for which protection was sought. The board stated that a skilled reader of the cited document had no reason to exclude the range of 85 to 115°C claimed in the patent in suit when carrying out the invention disclosed in the citation. The teaching of the cited document was clearly not limited to the use of the exemplified temperatures, but extended to the whole described temperature range of 80 to 170°C which had been made available to the skilled person as a technical teaching. The subject-matter of the patent in suit lacked novelty.

In **T 406/94** the board found that the percentage range cited in the prior art, although numerically close to the claimed range, could not be adduced to anticipate the subject-matter claimed, because the percentage cited in the prior art was based on different starting materials.

In **T 610/96** the patentee claimed a magnetoresistive material comprising magnetic and non-magnetic metallic thin film layers. The board found that the claimed ranges defining the composition of these layers must be considered as a narrow selection of the generic disclosure of prior art document D10, which did not overlap with the sub-ranges preferred in D10 and which further selected a specific non-magnetic layer among a group of possible layers. This selection was also sufficiently far removed from the specific examples of D10. Furthermore, the claimed material showed different characteristics of the magnetoresistance change, so that the specific sub-range was not simply an arbitrary part of the generic disclosure of D10, but was of a different nature and therefore novel. The criteria for selection inventions set out in **T 279/89** were thus satisfied. In **T 100/12** too, all three criteria were met.

In **T 230/07** the board noted that novelty and inventive step are two distinct requirements for the patentability of an invention and therefore different criteria should apply for their assessment. So, the presence or absence of a technical effect within a sub-range of numerical values was not to be taken into account in the assessment of novelty. To establish novelty of a sub-range of numerical values from a broader range, the selected sub-range should be narrow and sufficiently far removed from the known broader range illustrated by means of examples. A sub-range is **not rendered novel by virtue of a newly discovered effect** occurring within it.

In **T 1130/09** the "selection invention" principle had been applied in the contested decision for the purpose of assessing novelty, regard being had to the three criteria developed in **T 198/84** (see OJ 1985, 209). The board observed that this principle was applicable where a narrow sub-range was selected from a broader range. The passage on page 9, lines 5 to 7, of document (2) disclosed that the dimensions of the structures were measured in nanometres or micrometres. Therefore, as had already been established in the contested decision, the range specifically claimed was a narrow selection which, in the absence of examples in document (2), had to be considered far removed from the central embodiments in that document. The board thus held that the first two of the criteria defined in **T 198/84** were met. The **third** – that a technical effect of the narrower sub-range claimed had to be demonstrated – could not, however, be considered for the purpose assessing novelty, because novelty and inventive step were two distinct requirements for patentability. A technical effect within the more narrowly claimed range did not confer novelty on a numerical range which was already novel per se, but merely confirmed its already established novelty. Whether or not there was a technical effect nevertheless remained **a matter of inventive step** (many decisions have applied this approach; see **T 1233/05**, **T 1131/06**, **T 230/07**, **T 913/07**, **T 2041/09**, **T 492/10**, **T 1948/10**, **T 423/12**, **T 378/12**, **T 1404/14** and **T 261/15**).

Indeed in **T 378/12**, concerning criterion (c), the board was of the view that purposive selection was relevant for assessing inventive step but not novelty. Consequently, since